A Multi-Agent Retrieval-Augmented Framework for Semantic Job Matching and Iterative Resume Optimization

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***Abstract*—In today's time, candidates use different applications for job applying. Most of these applications work on keyword matching that is, matching the words in user's resume and job description, and the job description with most matching words is shortlisted. But this match purely depends on text being exactly same but sometimes some job roles requirements which are same are mentioned in different ways. Here, the keyword matching fails and which leads to jobs not getting shortlisted despite having high chance of candidate getting selected for the job. On the other hand, resumes in the real world are created once by a candidate and used at many places, but different jobs require showing the same resume in a different way by using more words from job description and tweaking the same resume. To tackle these problems, we have created a unified approach called "PlaceMates" that uses a semantic similarity matching feature to get the actual meaning of both job requirements and user profiles, which then gives context to other successive models for the generation of job-specific resumes, and optimization of the resume occurs through multiple rounds. It starts with representing candidate profiles and job descriptions as embeddings to get meaning beyond the keyword approach. Now, this candidate profile is compared with job description to find relevant jobs. For each selected job, resume generation process begin. It starts with drafting good resumes which is relevant to job description and candidate profile, this resume is taken as reference. Resume optimization begins with taking that resume as guidance, over multiple iteration resume keeps improving. The results shows that understanding meaning and using good reference example helps resume to improve over time iteratively and better aligned with job requirements.**

I. Introduction

Discovering a job in today's time occurs at different level. It includes resume generation from manual user input and refining that with generic suggestions, and job matching often occurs by keyword-based filtering of that resume. There is no platform where user profiles and job matching are done simultaneously by a semantic approach that is focuses on actual meaning rather than text. When keyword-based filtering is used, some of high chance job get filtered out just because in the user's resume that type of word is not used. Using a different resume for different job roles creates a positive impact on the chance of getting selected, but this often needs manual tweaking of the resume each time, which many candidate do not do instead they use the same resume for applying at different roles. This leads to candidates who are strong but with a generic single resume using for different roles get rejected because different roles requires differently presenting the same skill set and work.

At the same time, recent progress in learning right way of representing data and use of large language models for improving content quality has started to change. Dense vector embeddings allow systems to go beyond simple keyword matching and capture deeper relationships between candidate profiles and job requirements [1]. And with retrieval-augmented generation (RAG), systems can bring in relevant information during generation, which makes the output not just fluent, but also more grounded and better aligned with

the task [2].

However, most of the existing system still apply this technique separately, like for recommending a job and for improving the quality of a resume without using both of them simultaneously. So there is a lack of a system that uses semantic matching, and using that representation of the candidate improves after multiples attempts of improvements

In this work, we propose *PlaceMates*, an approach where we try to bridge this gap by combining both semantic matching of job and user profile and improving user profile representation through multiple rounds of refinements. Our approach starts with collecting data from multiple sources and representing them in a vector database. The system also collects jobs and job descriptions posted on multiple platforms and this is also converted into vector embeddings. It then perform vector similarity search, which helps find similarity beyond the keyword that is, similarity based on actual meaning. Through this, we get a high score job for the user profile. Now, our system starts, for each job, our system first drafts a high-score resume, which is present in our database, and by taking this as a reference, our multi-agent mechanism starts improving the resume. The resume content gets evaluated at each round, and over time resume get improved as per the target role's requirements.

The main contributions of this work are summarized as follows:

- A unified framework that brings together semantic job matching and dynamic candidate representation within a single end-to-end pipeline.
- The use of a multi-agent, feedback-driven mechanism to iteratively refine job-specific resume content.
- A semantic matching approach built on dense embeddings, which helps capture context beyond traditional keyword-based methods.
- An evaluation strategy that continuously monitors the refinement process and guides it in a structured manner.
- A self-improving data pipeline that enhances retrieval and generation performance over time

II. Related Work

A. Traditional Job Matching Systems

Early approaches to job matching relied on keyword-based techniques, where candidate profiles and job descriptions were compared using term frequency and lexical overlap. Methods such as TF-IDF and rule-based filtering enabled efficient matching but lacked the ability to capture semantic relationships between skills and requirements. Consequently, these systems often fail to identify relevant matches when similar concepts are expressed using different terminology. While later works introduced machine learning-based recommendation methods, many still depend on surface-level representations and do not fully address contextual alignment between candidates and job roles [3]–[5].

B. Semantic Matching and Dense Retrieval

To address these limitations, recent research has shifted toward semantic matching using dense vector representations. Sentence embedding models map text into continuous vector spaces, enabling comparison based on contextual similarity rather than exact term overlap [1]. Techniques such as contrastive learning further improve embedding quality by aligning semantically similar inputs [6]. Efficient retrieval in high-dimensional spaces is supported by vector indexing methods such as FAISS, which enable scalable nearest neighbor search [7]. These methods are further optimized using techniques such as product quantization for efficient similarity computation [8]. Benchmark studies such as BEIR demonstrate the effectiveness of dense retrieval across diverse tasks [9]. While these approaches significantly improve retrieval quality, they are typically applied independently of downstream content generation and adaptation tasks.

C. Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) combines information retrieval with generative modeling by incorporating relevant external context into the generation process. This approach improves factual grounding and contextual relevance of outputs [2]. Several extensions have explored improved retrieval strategies and integration techniques to enhance generation quality [10]–[12]. Although RAG has demonstrated strong performance in knowledge-intensive tasks, its application to job-specific content generation and dynamic candidate profiling remains limited.

D. Multi-Agent Language Model Systems

Recent work in large language models has introduced multi-agent frameworks that enable iterative reasoning and refinement. These systems use interacting components to generate, evaluate, and improve outputs over multiple iterations. Approaches such as ReAct integrate reasoning with action [13], while methods like Self-Refine and Reflexion employ feedback-driven refinement mechanisms [14], [15]. These methods demonstrate the effectiveness of iterative optimization; however, their integration into structured, domain-specific pipelines such as resume generation is still underexplored.

E. Self-Improving and Adaptive Systems

Learning from feedback has become central to modern AI systems. Techniques such as reinforcement learning from human feedback (RLHF) improve model alignment through iterative optimization [16], while approaches like Constitutional AI introduce structured feedback mechanisms to guide behavior [17]. While these approaches focus on improving model behavior, the use of continuously evolving datasets to enhance retrieval and generation within application-specific pipelines remains relatively unexplored.

F. Resume Processing and Screening Systems

Automated resume processing systems have been widely studied to extract structured information from unstructured documents. Techniques based on natural language processing enable parsing of resumes into structured representations of skills, experience, and education [18]. Additionally, automated screening systems are commonly used in recruitment pipelines to filter candidates based on predefined criteria [19].

However, such systems often rely on rigid rule-based or keyword-driven approaches, limiting their ability to capture deeper contextual relationships between candidate profiles and job requirements. As a result, they may overlook relevant candidates when skills are expressed in different forms. Moreover, concerns regarding bias and lack of transparency in automated hiring systems have been highlighted in real-world deployments [20].

These limitations point to the need for more intelligent and context-aware approaches that do more than just process resumes. Ideally, such systems should also be able to generate and refine them based on the specific job requirements. This, in a way, motivates combining semantic understanding with adaptive generation techniques in the proposed system.

G. Research Gap

Even though there have been clear advancements in areas like semantic retrieval, generative modeling, and multi-agent systems, most existing approaches still operate somewhat independently. Job matching systems, for instance, mainly focus on retrieval and do not really adapt how a candidate is represented. On the other hand, resume generation tools often work without much connection to the actual job context. In addition, iterative refinement—where outputs are continuously improved—is not commonly part of end-to-end pipelines. Because of this, there is still a lack of systems that can dynamically adjust candidate representation based on job requirements while also bringing together matching, generation, and evaluation in a single unified framework. This gap is what motivates the approach proposed in this work.

III. Methodology

A. System Overview

The proposed system, *PlaceMates*, is designed as an end-to-end intelligent pipeline that integrates semantic job matching, retrieval-augmented resume generation, and iterative refinement. The objective is to transform raw candidate information into highly optimized, job-specific resumes aligned with target job requirements.

As illustrated in Figure. 1, the system follows a structured multi-stage pipeline. Initially, candidate data is collected and organized into a unified profile. This profile is then matched with relevant job opportunities using semantic similarity rather than traditional keyword matching, enabling more context-aware job selection.

Once relevant jobs are identified, the system generates tailored resumes using a retrieval-augmented approach. Instead of generating content in isolation, the system retrieves high-quality examples and injects them as context to guide generation. The generated resumes are then refined through an iterative feedback mechanism, ensuring continuous improvement in quality.

Finally, high-quality outputs are retained and reused in future iterations, forming a feedback-driven loop that enhances the system's performance over time. This design enables the system to not only generate resumes but also improve its effectiveness with usage, similar to adaptive learning approaches in modern AI systems [14], [15].

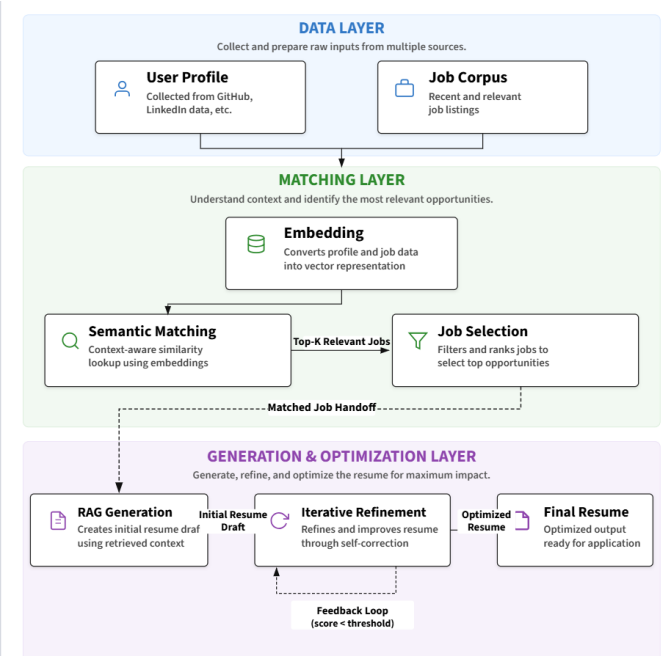


Fig. 1. End-to-end architecture of the proposed system illustrating multi-source ingestion, semantic matching, retrieval-augmented generation, and feedback-driven refinement.

B. Architecture Design

The architecture is organized into three functional layers: data ingestion, semantic matching, and generation with refinement.

Data Layer: The system collects user information from multiple sources such as GitHub and LinkedIn. This data is structured into a unified profile that captures skills, experience, and domain knowledge.

Matching Layer: Both candidate profiles and job descriptions are converted into vector embeddings, enabling comparison in a shared semantic space. This allows the system to identify relevant job opportunities based on contextual similarity rather than exact keyword matches.

Generation and Optimization Layer: For each selected job, the system generates a resume using retrieval-augmented generation and improves it through an iterative refinement loop. This ensures that the final output is both relevant and high quality.

This layered design ensures modularity, scalability, and clear separation of responsibilities across the system.

C. Semantic Job Matching

The semantic matching process aims to identify the most relevant job opportunities for a given candidate profile.

Both candidate data and job descriptions are converted into vector embeddings, which represent their meaning in a high-dimensional space. This allows the system to measure similarity based on context rather than exact keyword overlap.

A similarity scoring mechanism is then used to compare candidate profiles with job descriptions. The system retrieves the top-K most relevant jobs and applies filtering to remove less suitable options based on predefined criteria such as relevance and recency.

This approach improves matching accuracy by capturing deeper relationships between skills and job requirements, enabling more personalized and meaningful job recommendations.

D. Retrieval-Augmented Generation

The retrieval-augmented generation (RAG) process, illustrated in Figure. 2, is a key component of the system that enhances resume generation using external knowledge.

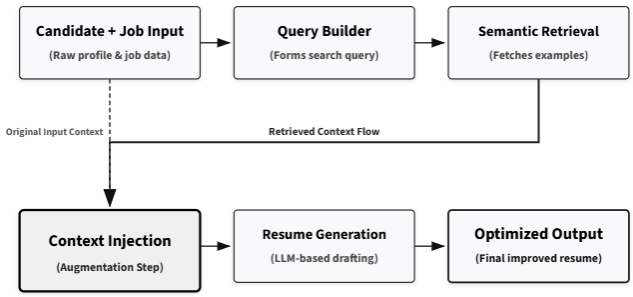


Fig. 2. Retrieval-augmented generation pipeline showing query formation, retrieval, and context-aware resume generation.

Typically, resumes are created directly from the input profile and the job description. In this approach, we first form a query by combining both user input and job description. This query is used to retrieve a relevant resume from our dataset. This dataset contain best quality resumes from different domain. So this resume are used as reference while writing new resumes, so with context meaning and keywords the system also have a reference of how a good resume should be, which make the generated resume with more structurally aligned. This approach improves content quality of generated resume by providing strong resumes so score of generated resume also becomes better than the baselines. In cases where system fails to fetch relevant examples then also system generate resume ensuring robustness. So in this way RAG helps this process by transforming the purely generative process to make it guided and context-aware

E. Multi-Agent Iterative Refinement

The system employs a multi-agent refinement mechanism to iteratively improve resume quality, as shown in Figure. 3.

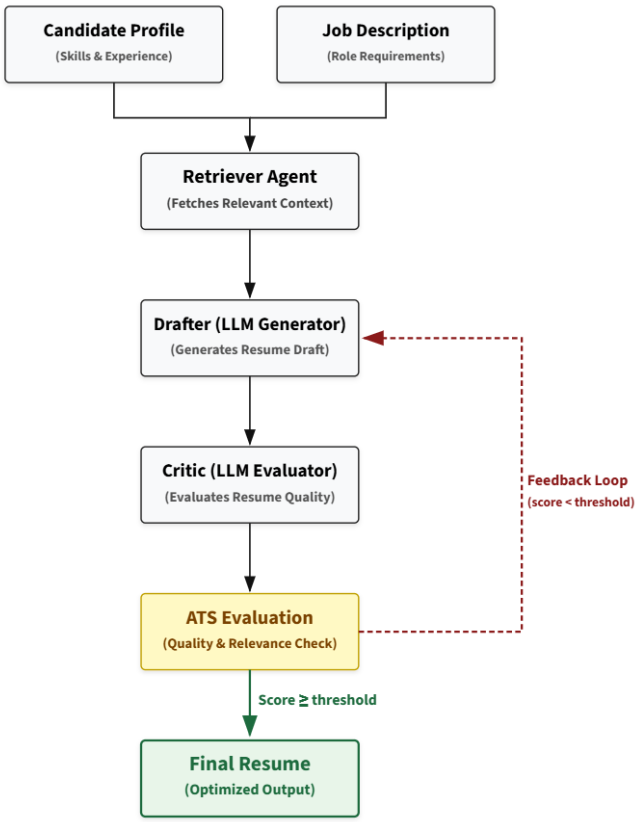


Fig. 3. Multi-agent iterative refinement loop illustrating interaction between retriever, drafter, and evaluator components.

This process consists of three main components:

- **Retriever:** Provides relevant contextual examples
- **Drafter:** Generates or updates the resume
- **Evaluator (Critic):** Assesses quality and provides feedback

The drafter generates an initial resume, which is then evaluated based on predefined quality criteria. The evaluator produces structured feedback highlighting areas of improvement.

This feedback is used in subsequent iterations to refine the resume. The process continues until a quality threshold is achieved or a maximum number of iterations is reached.

This iterative approach ensures:

- Continuous improvement in resume quality
- Better alignment with job requirements
- Reduced errors and inconsistencies

By separating generation and evaluation, the system avoids uncontrolled outputs and achieves more reliable refinement.

F. Evaluation Mechanism

The evaluation mechanism, shown in Figure. 4, is responsible for measuring resume quality and guiding the refinement process.

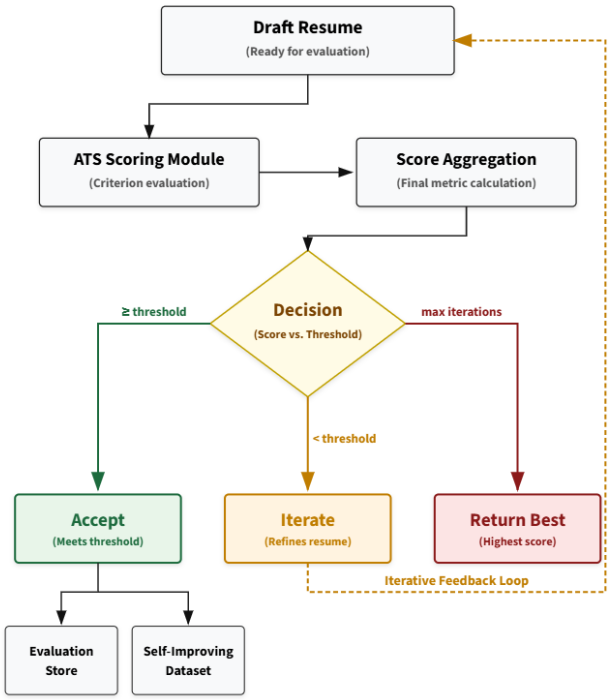


Fig. 4. Evaluation mechanism illustrating scoring, decision logic, and feedback-driven refinement.

The system computes a composite score based on multiple criteria, including:

- Relevance to job requirements
- Clarity and impact of content
- Structural completeness

This score approximates the behavior of Applicant Tracking Systems (ATS), which are commonly used in real-world hiring processes.

A decision step is used to check the score against a fixed threshold:

- If the score is equal to or higher than the threshold, the resume is accepted.
- If not, the system keeps improving the resume step by step.

To avoid too many repetitions, a maximum number of attempts is set. After all the attempts if the score still does not reach the threshold, the resume with the best score is selected. Overall, this approach make sure the quality should not comprimise and also makes sure the process does not run unnecessarily.

G. Self-Improving Dataset

The system contains a dataset of high-quality resumes, which are used as references for new resume generation. To ensure that this dataset does not remain static and improves over time, resumes with high scores above a quality threshold during evaluation are selected and stored in this dataset. Before adding them, the system makes sure that sensitive data is not included.

Because of this, the resumes keep getting better over time and do not become outdated by using old or poor references. In simple terms, the system keeps updating itself with better resumes, so it always has good examples to use when creating new ones.

IV. Results and Evaluation

A. Experimental Setup

The proposed system was evaluated on a dataset of 25 distinct user profiles spanning multiple technical domains, including web development, data science, and software engineering. The profiles were synthetically constructed to represent diverse skill distributions, experience levels, and role preferences, ensuring evaluation across varied real-world scenarios.

Each profile was matched against a continuously updated job corpus, resulting in 130 end-to-end pipeline executions. Each execution consists of semantic job matching, retrieval-augmented resume generation, and iterative refinement.

The evaluation focuses on three key aspects: (i) job matching relevance, (ii) resume quality, and (iii) behavior of iterative refinement across different user profiles.

B. Semantic Job Matching Performance

To evaluate job matching quality, three complementary metrics were used:

- **Precision@5**: measures how many of the top suggested jobs are actually relevant, indicating the usefulness of recommendations
- **Recall@5**: measures how many relevant jobs are successfully retrieved, reflecting coverage of opportunities
- **NDCG@5**: evaluates how well relevant jobs are ranked, giving higher importance to placing the most suitable jobs at the top

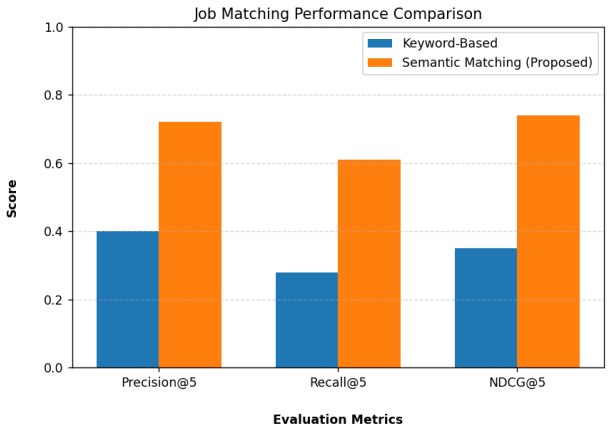


Fig. 5. Comparison of job matching performance across evaluation metrics

As shown in Figure. 5, the proposed semantic matching approach consistently outperforms keyword-based methods across all evaluation metrics. The improvement indicates that semantic representations enable better identification and prioritization of relevant job opportunities.

C. Resume Generation Quality

The quality of generated resumes was evaluated using an automated scoring mechanism designed to approximate applicant tracking systems (ATS).

TABLE I
RESUME QUALITY COMPARISON

Method	Average ATS Score
Base Resume	62
Single-Pass Generation	74
Proposed System	88

Table I shows that the proposed system achieves a substantial improvement in resume quality. The increase from 62 to 88 highlights the effectiveness of combining contextual retrieval with iterative refinement. The improvement over single-pass generation further demonstrates the importance of incorporating feedback into the generation process.

D. Iteration Behavior Analysis

To understand how resume quality evolves across iterations, representative user profiles were analyzed.

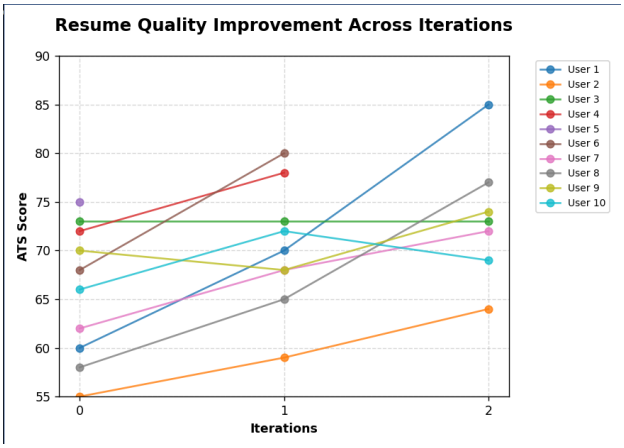


Fig. 6. Resume quality progression across iterations for different user profiles

As shown in Figure. 6, resume quality generally improves across iterations, although the behavior varies across users. Some profiles exhibit rapid improvement within a single iteration, while others demonstrate gradual gains. In certain cases, minimal change is observed, indicating that the initial draft already satisfies key requirements.

The results also indicate that resume quality does not always improve monotonically across iterations. In a few cases, minor fluctuations are observed. To ensure output reliability, the system retains the highest-quality version generated during the refinement process.

E. System Robustness

The system was evaluated under constrained conditions, including limited resource availability and partial execution failures. In such scenarios, fallback mechanisms enabled the pipeline to continue execution and produce valid outputs.

Across the evaluated cases, the system consistently generated at least one acceptable resume, demonstrating robustness and stability in practical deployment settings.

F. Ablation Study

An ablation study was conducted to assess the contribution of key system components.

TABLE II
ABLATION STUDY RESULTS

ConFigureuration	Average ATS Score
Without Retrieval (No RAG)	70
Without Iterative Refinement	65
Full System (Proposed)	88

The results show that removing retrieval significantly reduces performance, while removing iterative refinement leads to a further drop in quality. This confirms that both contextual grounding and iterative improvement are essential for achieving high-quality outputs.

G. Summary of Results

The experimental results demonstrate that the proposed system:

- Significantly improves job matching relevance through semantic understanding
- Enhances resume quality using retrieval-augmented generation
- Achieves consistent improvements across iterations with diverse behavioral patterns
- Maintains stable performance under practical constraints

Overall, the findings validate the effectiveness of the proposed unified framework for automated, job-specific resume optimization.

V. DISCUSSION

The results show that job matching and resume generation becomes more easier and efficient by semantic approach because it understand the actual meaning rather than keywords.

By using appropriate resumes from the database, the quality of the resume gets enhanced over multiple rounds. However, we noticed that different users had different results. Some got a high-quality resume almost immediately, while others faced small ups and downs before getting a high-quality resume.

In some cases, the resume quality might be degraded during a specific resume optimization round, resulting in a resume with a low ATS score. In such cases, the system automatically selects the resume version with the highest ATS score as the final resume.

This shows why it is important to check the resume at every stage instead of only the final version of the resume.

The platform is reliable and manages to produce better results even under different conditions.

However, as our testing was done with a small group of synthetic user profiles, real-world results might change depending on the data quality and the AI models used.

Ultimately, these results show that combining semantic matching with a continuous resume optimization loop is a practical way to automatically build high-quality resumes for specific jobs.

VI. Conclusion and Future Work

In this paper, we introduced *PlaceMates*, a unified platform that links semantic job matching with RAG-based resume generation with continuous optimization loop. By matching user profiles and job postings in the shared semantic space, the system can find better job recommendations for users rather than traditional keyword-based approaches.

Using RAG allows the system to generate job-specific resumes, while the resume optimization loop improves the quality of the resume over multiple rounds. Our findings show high improvement in job matching accuracy and resume quality, with consistent results across all the test users.

However, we noted some limitations. For current evaluation, testing was done with synthetic user profiles, so performance in real-world might vary. The final output also depends on how good retrieved resume examples are and how the AI models perform.

Going forward, we plan to focus on scaling the system with larger and more diverse datasets and including real user feedback into the loop. We also see a chance to improve our retrieval methods so that resumes are generated with the right context. Beyond this, we could personalize the system, which would align the system towards better individual needs and changing job market trends.

Overall, our approach suggests that linking semantic retrieval with an iterative feedback loop is a practical and effective way to automate job-specific resume generation.

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